

Agentic LLM-RAG Framework for Self-Improving Predictive Maintenance in Fleet Telematics

Literature Survey

1. LLM / RAG for Maintenance and Vehicle Fault Diagnosis

a) Prescriptive Agents based on RAG for Automated Maintenance (PARAM)

Year / Venue: 2025 (arXiv preprint, v2) | **Domain:** Industrial predictive / prescriptive maintenance (bearing vibration)

Authors: Harbola et al.

Achieved work: Converts numerical bearing-vibration telemetry into text so an LLM can do few-shot anomaly detection. A multi-agent RAG component searches maintenance manuals (vector embeddings, semantic search) and the web, and the LLM (Gemini) outputs structured recommendations: immediate actions, inspection checklist, corrective actions, parts and schedule.

Limitations:

- Validated on one bearing dataset, not on vehicles.
- Performance depends on the LLM and is untested across languages and manual formats.
- The system does not learn from the outcomes of its own recommendations; edge deployment is left open.

PDF link: <https://arxiv.org/pdf/2508.04714>

b) VBFD-Agent: EV Battery Fault Detection and Diagnosis with Descriptive Text Modeling

Year / Venue: 2026 (arXiv preprint) | **Domain:** Electric-vehicle battery fault diagnosis

Authors: Chan, Chen and Pan (Shanghai Jiao Tong University)

Achieved work: Turns battery signals (speed, voltage, current, SOC, temperatures, insulation resistance) into structured text. Retrieves similar historical cases, predicts alarm codes by similarity-weighted voting, retrieves manual chunks, and an LLM (DeepSeek-V3.2) outputs JSON diagnosis and actions. Tested on about 5.5 million real records from ten EVs and rated by three battery experts.

Limitations:

- Only ten vehicles and one component family.
- Training-free, so the case memory is built in advance and is not updated from outcomes.
- Depends on case coverage and knowledge-base quality; rare or safety-critical faults still need expert checking.

PDF link: <https://arxiv.org/pdf/2605.20742>

c) Multi-Agent Causal Reasoning System for Error Pattern Rule Automation in Vehicles (CAREP)

Year / Venue: 2026 (arXiv; ICLR 2026 workshop) | **Domain:** Automotive diagnostic trouble codes (DTC) and error-pattern rules

Authors: Math, Lorenz and Lienhart

Achieved work: Multi-agent system that generates Boolean error-pattern rules from DTC sequences by combining causal discovery with LLM agents. On a real dataset of 29,100 unique codes and 474 ground-truth rules it reconstructs rules more accurately than standard RAG baselines.

Limitations:

- Targets rule generation for manufacturer fleet monitoring, not the maintenance decision or scheduling step.
- No ongoing learning loop is described.

PDF link: <https://arxiv.org/pdf/2602.01155>

d) Our Cars Can Talk: How IoT Brings AI to Vehicles

Year / Venue: 2025 (arXiv preprint) | **Domain:** Connected vehicles, IoT and edge AI for predictive maintenance

Authors: Authors not captured; see PDF

Achieved work: Vision paper describing a pipeline: vehicle sensor data is cleaned and normalised, edge ML detects anomalies and predicts faults, and generative AI agents turn the fused context into conversational, proactive maintenance advice. States that edge AI plus LLMs for vehicle predictive maintenance is largely unexplored.

Limitations:

- Conceptual, with no experimental benchmark results.
- Focused on individual consumer vehicles rather than fleets.

PDF link: <https://arxiv.org/pdf/2507.17214>

e) Exploring LLM-based Frameworks for Fault Diagnosis

Year / Venue: 2025 (arXiv preprint) | **Domain:** Industrial time-series sensor fault detection

Authors: Hitachi America R&D team

Achieved work: Systematically compares single-LLM and multi-LLM systems and input forms (raw, statistical summary, hybrid) for fault detection, including in-context learning, explanations and continual learning. Finds that summarised statistical inputs work best and that several specialised LLMs improve fault-classification sensitivity.

Limitations:

- Industrial sensor data rather than vehicle fleets.
- No retrieval from manuals or repair history.

PDF link: <https://arxiv.org/pdf/2509.23113>

f) From Prediction to Prescription: LLM Agent for Context-Aware Maintenance Decision Support

Year / Venue: PHM Society European Conf. (confirm year on paper page) | **Domain:** Prescriptive maintenance planning (general systems)

Authors: Deng, Namoano, Zheng, Khan and Erkoyuncu (Cranfield University)

Achieved work: LLM agent that uses RAG and tool calling to turn machine-learning predictions into context-aware maintenance plans, using the system's maintenance documentation as its knowledge base.

Limitations:

- Not specific to vehicles.
- No evidence of learning from past maintenance decisions.

PDF link: <https://papers.phmsociety.org/index.php/phme/article/download/4114/2428>

g) Toward Autonomous LLM-Based AI Agents for Predictive Maintenance: State of the Art, Challenges, and Future Directions

Year / Venue: 2025 (Applied Sciences 15(21), 11515) | **Domain:** Predictive maintenance review: discriminative, generative and LLM-agent approaches

Authors: Politecnico di Torino authors (see paper)

Achieved work: Reviews discriminative, generative and LLM-based approaches to predictive maintenance and consolidates the fragmented evidence on LLM-driven agents (perception, reasoning, action across the PdM lifecycle). Proposes a unified architectural view for maintenance workflows.

Limitations:

- Review paper with no new experiments.
- Not focused on fleet telematics or on outcome-based self-improvement.

PDF link: <https://iris.polito.it/retrieve/handle/11583/3005359/948607>

h) AI-Driven Predictive Maintenance with Environmental Context Integration for Connected Vehicles: Simulation, Benchmarking, and Field Validation

Year / Venue: 2026 (arXiv preprint) | **Domain:** Connected-vehicle / fleet predictive maintenance

Authors: Khemani et al.

Achieved work: Contextual data-fusion framework that combines in-vehicle sensor streams with road quality, weather, traffic and driver behaviour obtained through V2X and third-party APIs, with inference at the vehicle edge (LightGBM, SHAP). Per-vehicle fine-tuning cut mean error on wear-driven events from 25.1 days to 12.2 days; validated in simulation and in the field.

Limitations:

- Uses classical ML rather than LLM reasoning, so it gives predictions without retrieval over manuals or repair history, and without explained recommendations.
- The wear-event ablation is small (6 events).

PDF link: <https://arxiv.org/pdf/2603.13343>

i) Sensor-Level Fault Diagnosis for Automotive Software Validation Using Large Language Models

Year / Venue: 2026 (arXiv preprint) | **Domain:** Automotive sensor fault diagnosis, ISO 26262 validation

Authors: See arXiv listing (author names not retrieved)

Achieved work: Converts windowed sensor signals into structured natural-language descriptors and embeds LLM inspection in a two-phase ISO 26262 workflow, so only requirement-violating recordings are diagnosed. Evaluates several open-source LLMs under LoRA fine-tuning against strong deep-learning baselines, producing a fault label and fault location.

Limitations:

- Test-bench (hardware-in-the-loop style) validation, not live fleet operation.
- Diagnoses faults rather than predicting failures ahead of time or scheduling maintenance.
- As far as the abstract shows, no manual retrieval or learning from outcomes.

PDF link: <https://arxiv.org/pdf/2608.05921>

j) Agent-based Condition Monitoring Assistance with Multimodal Industrial Database RAG (MindRAG)

Year / Venue: 2025 (arXiv preprint) | **Domain:** Industrial condition monitoring (rotating machinery, bearings)

Authors: Lowenmark, Strombergsson, Liu, Liwicki and Sandin

Achieved work: Modular framework (MindRAG) that combines multimodal RAG with new vector-store structures designed for condition-monitoring data, plus reasoning agents that answer real condition-monitoring queries, with an experimental framework for evaluating such agents in realistic industrial scenarios.

Limitations:

- Process-industry machinery rather than vehicles or fleets.
- Not tested on telematics data or on missing and delayed signals.

PDF link: <https://arxiv.org/pdf/2506.09247>

2. IEEE Papers: V2X Telemetry, RAG and Continual Learning

a) LLM-Based Telemetry Repair and Fault Detection in V2X Networks with Digital Twin Guidance

Year / Venue: 2025 (IEEE GLOBECOM 2025, Taipei) | **Domain:** Vehicle-to-everything (V2X) telemetry, fault detection

Authors: Hazarika, Singh, Canberk and Duong

Achieved work: Framework that uses LLMs guided by digital-twin context and RAG with environmental priors to recover high-dimensional telemetry lost during wireless outages. Adds federated continual learning to keep fault-classification performance across non-i.i.d. V2X conditions without centralised data exchange.

Limitations:

- Network-level telemetry recovery, not vehicle component maintenance.
- No maintenance recommendation or repair-outcome feedback.
- Conference-length paper; the PDF sits behind IEEE access.

PDF link: <https://research.itu.edu.tr/en/publications/llm-based-telemetry-repair-and-fault-detection-in-v2x-networks-wi/>

b) V2LLM: A Digital-Twin-Aware Retrieval-Augmented LLM for Resilient Telemetry Recovery and Fault Diagnosis in V2X

Year / Venue: 2025-26 (IEEE Trans. Network Science and Engineering; confirm year) | **Domain:** V2X telemetry recovery and continual fault diagnosis

Authors: Hazarika, Singh, Vo, Canberk, Cotton, Shin and Duong

Achieved work: Extends the GLOBECOM work into a journal framework: token-efficient telemetry reconstruction using local history, memory-guided context and digital-twin predictions, with a sliding-window autoregressive scheme for long gaps and a federated continual-learning module that adapts fault classifiers at roadside units without raw data sharing.

Limitations:

- Targets wireless V2X networks, not fleet maintenance decisions.
- Relies on a digital twin, and does not recommend maintenance actions or learn from repair outcomes.
- Full PDF needs IEEE access.

PDF link: <https://research.itu.edu.tr/en/publications/v2llm-a-digital-twin-aware-retrieval-augmented-large-language-mod/>

3. Agentic RAG and Self-Improving Agents

a) Agentic Retrieval-Augmented Generation: A Survey on Agentic RAG

Year / Venue: 2025 (arXiv survey) | **Domain:** Agentic RAG (general)

Authors: Singh, Ehtesham, Kumar and Khoei

Achieved work: Defines Agentic RAG as RAG in which autonomous agents manage retrieval through reflection, planning, tool use and multi-agent collaboration. Gives a taxonomy (single-agent, multi-agent, hierarchical, corrective, adaptive, graph-based) and compares it with naive and modular RAG.

Limitations:

- Survey with no experiments of its own.
- Does not cover predictive maintenance or telematics.

PDF link: <https://arxiv.org/pdf/2501.09136>

b) A Survey of Self-Evolving Agents: What, When, How, and Where to Evolve

Year / Venue: 2025 (arXiv survey) | **Domain:** Self-improving / self-evolving LLM agents

Authors: Gao et al.

Achieved work: Organises self-evolution around what to evolve (model, memory, tools, architecture), when (within or between tasks) and how (scalar rewards, textual feedback, single or multi-agent). Also covers evaluation, applications and open challenges.

Limitations:

- Broad survey with little on industrial or vehicle settings.
- Limited guidance on safe self-improvement for safety-relevant decisions.

PDF link: <https://arxiv.org/pdf/2507.21046>

c) Reflexion: Language Agents with Verbal Reinforcement Learning

Year / Venue: 2023 (NeurIPS 2023) | **Domain:** Self-improving language agents

Authors: Shinn et al.

Achieved work: After a failed attempt the agent writes a short verbal reflection on what went wrong, stores it in memory and reads it on the next attempt. This improves decision-making, reasoning and coding performance without retraining.

Limitations:

- Depends on the model diagnosing its own mistakes correctly.
- Small memory buffer, and tested on single agents in text-based tasks, not on sensor data.

PDF link: <https://arxiv.org/pdf/2303.11366>

d) Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection

Year / Venue: 2023 (arXiv); ICLR 2024 | **Domain:** Adaptive / self-reflective RAG

Authors: Asai, Wu, Wang, Sil and Hajishirzi

Achieved work: Trains a language model to decide when to retrieve and to critique both the retrieved passages and its own output using special reflection tokens. The 7B and 13B models outperformed ChatGPT and retrieval-augmented Llama-2-chat on open-domain QA, reasoning and fact-verification, with better citation accuracy.

Limitations:

- Needs extra training of the critic and generator, and adds inference overhead.
- Evaluated on text QA and generation benchmarks, not on sensor, telematics or maintenance data.

PDF link: <https://arxiv.org/pdf/2310.11511>

e) Voyager: An Open-Ended Embodied Agent with Large Language Models

Year / Venue: 2023 (arXiv) | **Domain:** Lifelong-learning LLM agent (Minecraft)

Authors: Wang et al.

Achieved work: LLM-powered agent that keeps learning through an automatic curriculum, a growing skill library of executable code, and iterative prompting with environment feedback and self-verification. Collected far more unique items and reached tech-tree milestones much faster than earlier methods, and reused its skills in new worlds.

Limitations:

- Tested in a simulated game, not a safety-critical setting.
- Relies on GPT-4, and generated skills can be wrong.
- Success is easy to verify in a game but outcomes are harder to judge in maintenance.

PDF link: <https://arxiv.org/pdf/2305.16291>

f) A-MEM: Agentic Memory for LLM Agents

Year / Venue: 2025 (arXiv) | **Domain:** Agent memory systems

Authors: Xu et al.

Achieved work: Zettelkasten-style memory in which each new memory becomes a note with attributes and links to related notes, and old notes are updated as new ones arrive (memory evolution). Improved results over baseline memory methods across several foundation models on long-term conversation benchmarks.

Limitations:

- Evaluated on conversational benchmarks, not industrial or time-series settings.
- Link quality depends on the LLM, and there is no ground-truth outcome feedback to confirm what is stored.

PDF link: <https://arxiv.org/pdf/2502.12110>

4. Small Language Models and Efficient Fine-Tuning

a) Small Language Models are the Future of Agentic AI

Year / Venue: 2025 (arXiv position paper) | **Domain:** Small language models for agents

Authors: Belcak et al. (NVIDIA Research)

Achieved work: Argues that small language models are capable enough, better suited and cheaper for many repetitive specialised agent tasks, and proposes small-by-default systems that call a large model only when needed, plus a method for converting agents.

Limitations:

- Position paper with limited experimental evidence, not peer-reviewed.
- Not about vehicles, telematics or connectivity limits.

PDF link: <https://arxiv.org/pdf/2506.02153>

b) QLoRA: Efficient Finetuning of Quantized LLMs

Year / Venue: 2023 (NeurIPS 2023) | **Domain:** Parameter-efficient fine-tuning

Authors: Dettmers, Pagnoni, Holtzman and Zettlemoyer

Achieved work: Fine-tunes a 4-bit quantised model through LoRA adapters, using the NF4 data type, double quantisation andpaged optimisers. Makes it possible to fine-tune a 65B model on a single 48 GB GPU without losing 16-bit fine-tuning quality; the Guanaco models reached close to ChatGPT level on the Vicuna benchmark.

Limitations:

- Evaluation centres on instruction-following and chat benchmarks, not on maintenance reasoning or telemetry.
- Benchmark-based chatbot evaluation has its own reliability limits.

PDF link: <https://arxiv.org/pdf/2305.14314>

Summary Table: Research Papers on Agentic LLM-RAG, Predictive Maintenance, Self-Improving Agents and Fleet Telematics (2023–2026)

S.No	Category	Paper	Year	Domain	Achieved work
1	LLM / RAG for Maintenance and Vehicle Fault Diagnosis	Prescriptive Agents based on RAG for Automated Maintenance (PARAM)	2025 (arXiv preprint, v2)	Industrial predictive / prescriptive maintenance (bearing vibration)	Converts numerical bearing-vibration telemetry into text so an LLM can do few-shot anomaly detection. A multi-agent RAG component searches maintenance manuals (vector embeddings, semantic search) and the web, and the LLM (Gemini) outputs structured recommendations: immediate actions, inspection checklist, corrective actions, parts and schedule.
2	LLM / RAG for Maintenance and Vehicle Fault Diagnosis	VBFDD-Agent: EV Battery Fault Detection and Diagnosis with Descriptive Text Modeling	2026 (arXiv preprint)	Electric-vehicle battery fault diagnosis	Turns battery signals (speed, voltage, current, SOC, temperatures, insulation resistance) into structured text. Retrieves similar historical cases, predicts alarm codes by similarity-weighted voting, retrieves manual chunks, and an LLM (DeepSeek-V3.2) outputs JSON diagnosis and actions. Tested on about 5.5 million real records from ten EVs and rated by three battery experts.
3	LLM / RAG for Maintenance and Vehicle Fault Diagnosis	Multi-Agent Causal Reasoning System for Error Pattern Rule Automation in Vehicles (CAREP)	2026 (arXiv; ICLR 2026 workshop)	Automotive diagnostic trouble codes (DTC) and error-pattern rules	Multi-agent system that generates Boolean error-pattern rules from DTC sequences by combining causal discovery with LLM agents. On a real dataset of 29,100 unique codes and 474 ground-truth rules it reconstructs rules more accurately than standard RAG baselines.
4	LLM / RAG for Maintenance and Vehicle Fault Diagnosis	Our Cars Can Talk: How IoT Brings AI to Vehicles	2025 (arXiv preprint)	Connected vehicles, IoT and edge AI for predictive maintenance	Vision paper describing a pipeline: vehicle sensor data is cleaned and normalised, edge ML detects anomalies and predicts faults, and generative AI agents turn the fused context into conversational, proactive maintenance advice. States that edge AI plus LLMs for vehicle predictive maintenance is largely unexplored.
5	LLM / RAG for Maintenance and Vehicle Fault Diagnosis	Exploring LLM-based Frameworks for Fault Diagnosis	2025 (arXiv preprint)	Industrial time-series sensor fault detection	Systematically compares single-LLM and multi-LLM systems and input forms (raw, statistical summary, hybrid) for fault detection, including in-context learning, explanations and continual learning. Finds that summarised statistical inputs work best and that several specialised LLMs improve fault-classification sensitivity.

S.No	Category	Paper	Year	Domain	Achieved work
6	LLM / RAG for Maintenance and Vehicle Fault Diagnosis	From Prediction to Prescription: LLM Agent for Context-Aware Maintenance Decision Support	PHM Society European Conf. (confirm year on paper page)	Prescriptive maintenance planning (general systems)	LLM agent that uses RAG and tool calling to turn machine-learning predictions into context-aware maintenance plans, using the system's maintenance documentation as its knowledge base.
7	LLM / RAG for Maintenance and Vehicle Fault Diagnosis	Toward Autonomous LLM-Based AI Agents for Predictive Maintenance: State of the Art, Challenges, and Future Directions	2025 (Applied Sciences 15(21), 11515)	Predictive maintenance review: discriminative, generative and LLM-agent approaches	Reviews discriminative, generative and LLM-based approaches to predictive maintenance and consolidates the fragmented evidence on LLM-driven agents (perception, reasoning, action across the PdM lifecycle). Proposes a unified architectural view for maintenance workflows.
8	LLM / RAG for Maintenance and Vehicle Fault Diagnosis	AI-Driven Predictive Maintenance with Environmental Context Integration for Connected Vehicles: Simulation, Benchmarking, and Field Validation	2026 (arXiv preprint)	Connected-vehicle / fleet predictive maintenance	Contextual data-fusion framework that combines in-vehicle sensor streams with road quality, weather, traffic and driver behaviour obtained through V2X and third-party APIs, with inference at the vehicle edge (LightGBM, SHAP). Per-vehicle fine-tuning cut mean error on wear-driven events from 25.1 days to 12.2 days; validated in simulation and in the field.
9	LLM / RAG for Maintenance and Vehicle Fault Diagnosis	Sensor-Level Fault Diagnosis for Automotive Software Validation Using Large Language Models	2026 (arXiv preprint)	Automotive sensor fault diagnosis, ISO 26262 validation	Converts windowed sensor signals into structured natural-language descriptors and embeds LLM inspection in a two-phase ISO 26262 workflow, so only requirement-violating recordings are diagnosed. Evaluates several open-source LLMs under LoRA fine-tuning against strong deep-learning baselines, producing a fault label and fault location.
10	LLM / RAG for Maintenance and Vehicle Fault Diagnosis	Agent-based Condition Monitoring Assistance with Multimodal Industrial Database RAG (MindRAG)	2025 (arXiv preprint)	Industrial condition monitoring (rotating machinery, bearings)	Modular framework (MindRAG) that combines multimodal RAG with new vector-store structures designed for condition-monitoring data, plus reasoning agents that answer real condition-monitoring queries, with an experimental framework for evaluating such agents in realistic industrial scenarios.
11	IEEE Papers: V2X Telemetry, RAG and Continual Learning	LLM-Based Telemetry Repair and Fault Detection in V2X Networks with Digital Twin Guidance	2025 (IEEE GLOBECOM 2025, Taipei)	Vehicle-to-everything (V2X) telemetry, fault detection	Framework that uses LLMs guided by digital-twin context and RAG with environmental priors to recover high-dimensional telemetry lost during wireless outages. Adds federated continual learning to keep fault-classification performance across non-i.i.d. V2X conditions without centralised data exchange.

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12	IEEE Papers: V2X Telemetry, RAG and Continual Learning	V2LLM: A Digital-Twin-Aware Retrieval-Augmented LLM for Resilient Telemetry Recovery and Fault Diagnosis in V2X	2025-26 (IEEE Trans. Network Science and Engineering; confirm year)	V2X telemetry recovery and continual fault diagnosis	Extends the GLOBECOM work into a journal framework: token-efficient telemetry reconstruction using local history, memory-guided context and digital-twin predictions, with a sliding-window autoregressive scheme for long gaps and a federated continual-learning module that adapts fault classifiers at roadside units without raw data sharing.
13	Agentic RAG and Self-Improving Agents	Agentic Retrieval-Augmented Generation: A Survey on Agentic RAG	2025 (arXiv survey)	Agentic RAG (general)	Defines Agentic RAG as RAG in which autonomous agents manage retrieval through reflection, planning, tool use and multi-agent collaboration. Gives a taxonomy (single-agent, multi-agent, hierarchical, corrective, adaptive, graph-based) and compares it with naive and modular RAG.
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Research Gaps Seen Across the 20 Papers

- **Few fleet-scale studies.** Vehicle papers use ten EVs, one component family or a test bench; PARAM, MindRAG and Hitachi use industrial machines.
- **Static knowledge bases.** VBFDD-Agent, PARAM and the maintenance RAG papers build their case memory or manual store in advance and do not update it from confirmed repair outcomes.
- **Self-improvement is shown only outside maintenance.** Reflexion, Voyager, Self-RAG and A-MEM learn from feedback, but in games, QA or chat, not in safety-relevant maintenance decisions.
- **Imperfect telematics data is rarely tested with LLM agents.** The V2X IEEE papers handle lost telemetry but do not produce maintenance recommendations.
- **Small versus large models is rarely measured on this task.** The SLM position paper and QLoRA motivate fine-tuned small models, but cost, latency and accuracy have not been compared for fleet maintenance.
- **Prediction and prescription are split.** Khemani et al. predict well with classical ML; Deng et al. and PARAM prescribe actions; no paper joins both on telematics with a learning loop.